AgroTwin-Share: Technical Master Guide

IoT & Hydro-Thermal Digital Twin Platform for Post-Harvest Produce Preservation

Document Purpose: Comprehensive Viva Defense, Project Expo Presentation & Technical Knowledge Transfer Document
Target Audience: Project Teammates, Technical Evaluators, Industry Judges & Viva Examiners
Repository: github.com/aaronaji007/AgroTwin | **Hosting:** Vercel Production Ready

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1. Executive Summary & Problem Context

The Food and Agriculture Organization (FAO) reports that approximately **33% to 45% of all harvested fruits and vegetables spoil before reaching consumers**, generating enormous economic losses ($1+ trillion globally) and squandering critical water and land resources. In cold-storage warehouses, legacy SCADA systems rely strictly on *passive threshold sensors*: they sound an alarm only after a temperature limit has been breached, with zero biological foresight into how long produce will actually remain marketable.

AgroTwin-Share bridges this technological divide by introducing a **Biophysical Digital Twin**. Instead of merely logging temperature and humidity, AgroTwin-Share continuously runs real-time mathematical models of cellular respiration kinetics (Arrhenius Q10 law), transpiration desiccation (Tetens Vapor Pressure Deficit), and hormonal climacteric ripening (Ethylene gas reaction cascades) to compute a dynamic **Shelf-Life Index (SLI %)** and exact days-to-spoilage countdowns for every storage bin in real-time.

> **[VIVA MASTER TIP] The 15-Second Pitch for Expo Judges**
>
> "AgroTwin-Share is an industrial Digital Twin platform for post-harvest cold storage. Rather than just recording room temperature, our platform simulates the living biology inside each produce batch—predicting cellular moisture loss, respiration decay, and ethylene ripening surges to dynamically forecast remaining shelf life in days, alerting facility operators before irreversible spoilage occurs."

2. Core Scientific & Mathematical Formulations

The foundational strength of AgroTwin-Share lies in its biophysical decay engine. The overall produce decay multiplier **K_decay** is calculated as a composite product of three independent environmental stress factors:

```
K_decay = K_base x f_thermal(T) x f_hydro(T, RH) x f_ethylene([C2H4])
where K_base is nominal daily decay rate at biological optimum setpoints.
```

2.1 Thermal Factor & Arrhenius Respiration (Q10 Law)

Fresh produce respires post-harvest, consuming stored sugars and oxygen while releasing CO2 and heat. According to Van 't Hoff's rule and the Arrhenius kinetics equation, the respiration rate roughly doubles to triples for every 10 deg C rise in temperature above optimal storage setpoints:

- **Above Optimal (T >= T_opt):** f_thermal(T) = Q10 ^ ((T - T_opt) / 10)
- **Chilling Injury Penalty (T < T_chill):** Tropical and sub-tropical produce (bananas below 12 deg C, avocados below 3 deg C) suffer irreversible cell membrane phase transitions causing rapid browning and cellular collapse: f_thermal = 1.0 + 1.5 x |T - T_chill|.

2.2 Hydro-Deficit & Transpiration Loss (Tetens VPD Model)

Relative humidity (RH %) alone is deceptive because warmer air holds significantly more water vapor. The true driving force for water evaporation from produce tissue into the chamber air is the **Vapor Pressure Deficit (VPD)** in kPa. We compute saturation vapor pressure **VP_sat** using the standard atmospheric Tetens equation:

- **Saturation Vapor Pressure:** VP_sat(T) = 0.61078 x exp((17.27 x T) / (T + 237.3)) [kPa]
- **Vapor Pressure Deficit:** VPD = VP_sat(T) x (1 - RH / 100)
- **Transpiration Stress Multiplier:** If VPD > VPD_opt, f_hydro = 1.0 + 1.8 x (VPD - VPD_opt). If RH > 96%, condensation occurs on produce skin, multiplying fungal/mold risk (Botrytis cinerea).

2.3 Phytohormone Ethylene Acceleration Factor (C2H4)

Ethylene is a natural ripening gas synthesized by **climacteric crops** (apples, bananas, avocados, tomatoes). Even trace levels (>0.10 ppm) trigger autocatalytic ripening genes that destroy chlorophyll, degrade pectin cell walls, and accelerate respiration:

- **Climacteric Acceleration:** f_ethylene = 1.0 + beta x ln(1 + 5 x ([C2H4] - Threshold))
- **Cross-Contamination Hazard:** Storing high-ethylene emitters (apples) alongside ethylene-sensitive non-climacteric crops (strawberries, greens) causes catastrophic early decay.

Table 1: Biological Commodity Profiles Programmed into AgroTwin-Share

Commodity	Type	Opt. Temp	Opt. RH	Q10	C2H4 Limit	Nominal Shelf-Life
Honeycrisp Apples	Climacteric	1.5 deg C	92%	2.2	0.15 ppm	180 Days (CA Storage)
Cavendish Bananas	Climacteric	13.5 deg C	90%	2.6	0.20 ppm	28 Days (Chill < 12 deg C)
Hass Avocados	Climacteric	5.0 deg C	90%	2.4	0.10 ppm	35 Days
Sweet Strawberries	Non-Climac.	0.5 deg C	93%	3.1	0.50 ppm	14 Days (Ultra perishable)
Roma Tomatoes	Climacteric	11.0 deg C	88%	2.3	0.30 ppm	24 Days

3. Full-Stack System Architecture & Engineering

AgroTwin-Share is architected as an enterprise-grade monorepo featuring a **Dual-Mode Service Layer**. This guarantees 100% operational autonomy during internet or server outages while providing seamless live integration with Python FastAPI microservices and cloud deployments.

Subsystem	Technology Stack	Key Architectural Responsibility
Frontend Dashboard	React 19, Vite, Tailwind CSS v4, Recharts, Lucide	Industrial SCADA user interface, real-time oscillating telemetry stream hook, interactive multi-axis oscilloscope charts, What-If projection simulation console.
Service Layer	Unified API Adapter (src/services/api.js)	Toggleable mock engine flag (const USE_MOCK = true). When true, in-memory client engine calculates all biophysical math. When false, seamlessly redirects to FastAPI.
Backend API	Python 3.12, FastAPI, Uvicorn, Pydantic v2	Modular RESTful API endpoints for storage bins, produce batch tracking, telemetry packet ingestion, historical logging, and Arrhenius decay projections.
Cloud & Edge	Vercel Serverless Functions, ASGI Handler (api/index.py)	Zero-config global deployment on Vercel with SPA client routing rewrites, sub-second cold starts, and automated continuous delivery.

3.1 The Toggleable Mock Service Layer & Dynamic Telemetry Hook

A critical engineering requirement fulfilled in src/services/api.js is the configuration toggle:

export const USE_MOCK = true;

When **USE_MOCK is true**, the client uses an internal Brownian fluctuation algorithm inside a React useEffect hook. Every 3 seconds, a timer oscillates chamber temperature, relative humidity, and ethylene concentration around realistic noise baselines. At every step, the digital twin recalculates the produce Shelf-Life Index (SLI %) so the dashboard is dynamic and active during live expo demonstrations. When flipped to **false** (or toggled via the header switch), all calls route to http://localhost:8000/api.

[VIVA QUESTION ALERT] Viva Question Alert: How does the toggleable service work?

"We decoupled our UI from our backend using a unified service adapter pattern in api.js. The frontend components make identical calls (fetchBins(), simulateEvent(), etc.). The adapter checks isMockEnabled(): if true, it reads from a thread-safe in-browser biophysical store; if false, it performs asynchronous fetch requests to FastAPI. This allows zero-latency offline expo demos with full production parity."

4. Comprehensive Module-by-Module Feature Walkthrough

The interface was redesigned with an Awwwards-caliber **Raw Industrial Editorial SCADA aesthetic**: high-contrast Swiss 1px borders, zero rounded corners, 60-30-10 color discipline, and Instrument Serif display numerals.

Module 01: Facility Control Matrix (Overview Page)

- **5 Macro SCADA Metric Cards:** Displays Active Units (5/5 Online), Mean Temperature (6.7 deg C), Mean Humidity (90.5%), Active Ethylene Spikes (Surge count), and Total Inventory Health ($193.5K valuation).
- **Chamber Telemetry Grid:** Real-time operational cards for Cold Room Alpha, Ripening Chamber Beta, CA Gamma, Pre-Cooling Delta, and Post-Harvest Epsilon with segmented SLI progress pips and days-to-spoilage countdowns.
- **Quick Mechanical Anomaly Injector:** One-click physical anomaly injection buttons (+C2H4 Spike, +Door Leak, Calibrate) to demonstrate instant real-time telemetry reaction to evaluators.

Module 02: Storage Chambers SCADA (Bin Detail View)

- **Chamber Switchboard Tabs:** Rapid mechanical switching between micro-zones (CR-ALPHA to PH-EPSILON).
- **Dual-Axis Oscilloscope Telemetry Chart:** Real-time Recharts plotting Temperature on Left Y-Axis and Humidity on Right Y-Axis with horizontal warning limit ReferenceLines.

- **Ethylene Concentration Curve:** Dedicated trace highlighting catalytic scrubber trip ceilings (e.g. 0.15 ppm).
- **SLI Decay Trajectory:** Projects biological degradation toward the commercial reject floor (50% SLI).
- **Physical Relay Switchboard:** Controls Chiller Compressor (4.8 kW inverter load), Circulation Fans (1,250 RPM), and Catalytic Ethylene Scrubbers (120 m3/h bed).

Module 03: Digital Twin Simulation Engine (What-If Panel)

- **Interactive Stress Sliders:** Live perturbation sliders for Temperature (-2 to 24 deg C), Relative Humidity (65 to 99%), Ethylene (0.01 to 2.0 ppm), and Projection Window (7 to 60 Days).
- **Comparative Trajectory Overlay:** High-contrast Recharts graph comparing Baseline Design vs. Simulated Perturbation.
- **Financial Exposure Ledger:** Instant calculation of dollar gain or loss on the batch (e.g. -$12,450 USD loss due to accelerated decay).
- **Biophysical Recommendation Directive:** Automated agronomic advice regarding priority dispatch or ventilation.

Module 04: Biomass Inventory Ledger (Batches View)

- **Lot Tracking Dossier:** Registry of all produce lots with variety classification (Highland Select Apples, Ecuador Gold Bananas, etc.).
- **Harvest & Intake Stamps:** Complete chain-of-custody tracking from field harvest to cold storage arrival.
- **Quality Grade Classification:** Export Grade A, Commercial Stage 3.5, and Direct Retail Ultra designations.

Module 05: Limits & Alarm Journal (Alert Drawer & Modal)

- **Configurable Safety Boundaries:** Operator customization of Min/Max Temp, Min/Max RH, and Ethylene Ceilings per chamber.
- **Commodity Quick Presets:** Instant setpoint calibration buttons: Apples CA (0.5-3.5 deg C), Banana Ripening (12.5-16 deg C), Avocado Pre-conditioning (4-7 deg C), Berry Hydro-chill (0-2.2 deg C).
- **Operator Acknowledgment Journal:** Slide-out drawer tracking unacknowledged alarms with timestamp and parameter values.

5. Step-by-Step Live Expo Demonstration Protocol

Follow this exact 3-minute demonstration sequence when presenting AgroTwin-Share to technical judges and expo visitors:

Step 1: Orient the Evaluator (30 Seconds)

Show the **Control Matrix** (Module 01). Explain the 60-30-10 industrial terminal aesthetic: "We designed this like an expensive, custom-built physical monitoring terminal. You are looking at Central Cold Storage Hub #4 holding $193,000 worth of perishable produce across 5 microclimate chambers." Point out the live 3-second telemetry pulse oscillating temperature and relative humidity.

Step 2: Trigger an Anomaly Event (45 Seconds)

Click the **[+C2H4 SPIKE]** or **[+DOOR LEAK]** button in the top right injector dock.
• Notice the instant state change: Ripening Chamber Beta turns to **[ALERT // WARNING]**.
• The alarm bracket in the header increments to **[ALARMS // 02]** with amber alert lighting.
• Open the **Storage Chambers** tab (Module 02). Show how the ethylene scrubber automatically switches to **[PURGING]** and the dual-axis oscilloscope graph plots the anomaly crossing the safety reference line.

Step 3: Demonstrate the What-If Digital Twin (60 Seconds)

Navigate to **Module 03 (Digital Twin Engine)**. Select *Honeycrisp Apples*.
• Move the Temperature slider up by 4 deg C. Watch the Recharts graph immediately update in real-time.
• Highlight the **Financial Exposure Ledger**: "By warming the chamber 4 deg C, the respiration rate doubles (Q10=2.2), cutting 65 days off shelf-life and causing an estimated $18,000 loss in batch asset value."
• Click **[RESET]** to show optimal baseline restoration.

Step 4: Demonstrate Full-Stack & Deployment Parity (45 Seconds)

Click the header toggle **[FASTAPI]**. Explain: "Our platform is full-stack. By switching this toggle, the frontend connects directly to our Python FastAPI REST server and Python digital twin engine running at port 8000. Furthermore, the entire project is deployed on Vercel with serverless API routes."

6. Top 15 Technical Viva Questions & Model Answers

Study these 15 questions thoroughly with your teammates before your viva examination. They cover theoretical, mathematical, architectural, and operational questions examiners love to ask:

Q1: What fundamentally differentiates your Digital Twin from a traditional IoT sensor dashboard?

A traditional IoT dashboard merely logs historical sensor values and applies simple static boolean thresholds (e.g. IF Temp > 4 deg C THEN Alarm). A Digital Twin, by definition, possesses an internal biophysical mathematical model that simulates the biological physiology of the produce. AgroTwin-Share computes enzymatic respiration kinetics, vapor pressure moisture loss, and ethylene gene triggering to forecast future degradation, yielding predictive shelf-life countdowns and dynamic economic loss models.

Q2: Why did you choose Vapor Pressure Deficit (VPD) instead of just using Relative Humidity (RH %)?

Relative humidity is a relative measure that depends heavily on air temperature: 90% RH at 2 deg C contains far less water vapor than 90% RH at 15 deg C. Produce skins experience moisture loss driven by the gradient between the internal saturated leaf/fruit tissue and the surrounding chamber air. Vapor Pressure Deficit (VPD in kPa) accurately measures this drying gradient using the Tetens equation. Lower VPD prevents desiccation, while too low VPD (RH > 96%) causes water droplets to condense, triggering Botrytis fungal mold.

Q3: What is the Q10 temperature coefficient and why is it important in post-harvest biology?

The Q10 temperature coefficient quantifies the rate of metabolic respiration increase when temperature increases by 10 deg C. For fresh horticultural crops, Q10 typically ranges from 2.0 to 3.5. For example, Honeycrisp Apples have a Q10 of 2.2; storing them at 11.5 deg C instead of 1.5 deg C causes them to burn through their stored sugars and organic acids at more than double the rate, halving their storage life.

Q4: What is Chilling Injury and how is it mathematically accounted for in your model?

Chilling injury is physiological damage that occurs at cold temperatures above freezing (typically 0 to 12 deg C depending on species). In subtropical crops like bananas (threshold 11.5 deg C) or avocados (3.0 deg C), cold causes cell membrane lipids to solidify from liquid-crystalline to gel phase, causing electrolyte leakage, browning, and failure to ripen. In our algorithm, if T < T_chill, a linear penalty multiplier (1.0 + 1.5 x delta_T) is applied to reflect accelerated collapse.

Q5: What is the difference between climacteric and non-climacteric produce in your platform?

Climacteric produce (bananas, apples, avocados, tomatoes) undergo an autocatalytic burst in respiration and ethylene synthesis at ripening onset. Non-climacteric produce (strawberries, citrus, leafy greens) do not exhibit this surge and do not ripen after harvest. In our engine, climacteric crops have high ethylene sensitivity slopes (beta = 1.8 to 3.2), meaning even trace ethylene triggers exponential decay cascades.

Q6: How does the platform calculate the Shelf-Life Index (SLI %)?

The Shelf-Life Index (SLI) is a normalized index (100% at harvest, 0% at total rot). Each time step delta_t, the drop in SLI is computed as: Delta_SLI = (BaseDailyDecay / 24) x K_decay x delta_t. When SLI drops below 75%, an action warning is logged; below 50%, produce has lost Grade A export status; below 10%, it is unmarketable waste. Days remaining is dynamically estimated as Current_SLI / (DailyDecay x K_decay).

Q7: How did you implement real-time live data without physical microcontrollers during the expo?

We engineered an autonomous client-side simulation layer in src/services/api.js and TelemetryContext.jsx. A dynamic React useEffect hook runs a 3-second interval applying subtle Brownian random walk noise to temperature (+-0.08 deg C), humidity (+-0.15%), and ethylene. It feeds these fluctuating inputs into the client-side biophysical model (digitalTwinModel.js), recalculating SLI in real time. This guarantees a lively, responsive dashboard with zero hardware dependencies.

Q8: How does switching between Mock Mode and FastAPI Live Mode work technically?

Our frontend API service exports a boolean flag 'const USE_MOCK = true;'. When true, all service functions return promises resolved from our in-browser mock store. When set to false (or switched via the UI toggle), the functions execute asynchronous HTTP fetch calls to the FastAPI backend ('http://localhost:8000/api/...'). The adapter normalizes Python snake_case keys into JavaScript camelCase, ensuring the React UI is completely agnostic to the data provider.

Q9: What REST endpoints does the FastAPI backend provide?

1. GET /api/summary (SCADA macro KPIs)
2. GET /api/bins and GET /api/bins/{id} (Chamber status and telemetry)
3. PUT /api/bins/{id}/thresholds (Safety limits configuration)
4. POST /api/bins/{id}/simulate-event (Hardware failure injection)
5. GET /api/batches (Biomass inventory lots)
6. GET /api/bins/{id}/history (Historical time-series telemetry with 1h, 6h, 24h, 7d ranges)
7. POST /api/telemetry (IoT sensor packet ingestion from field microcontrollers)
8. POST /api/simulate/sli (Digital twin What-If scenario projection calculations)
9. GET /api/alerts and POST /api/alerts/{id}/acknowledge (Alarm journal).

Q10: How is the financial economic loss calculated in the What-If simulation?

When an operator adjusts setpoints (e.g. higher temperature or lower humidity), the engine runs a day-by-day trajectory over the projection window (e.g. 14 days). It compares the simulated days remaining against the baseline days remaining. If the shelf-life delta is negative, the loss ratio is calculated as: Loss = Batch_Value_USD x (|Delta_Days| / Baseline_Days) x 0.75 (reflecting commercial discount and salvage value loss).

Q11: Why did you adopt an 'Industrial Editorial' design aesthetic instead of standard SaaS templates?

Standard SaaS templates with rounded corners, soft pastel drop-shadows, and bright violet icons look like consumer toys, not mission-critical industrial systems. We implemented an Awwwards-inspired Raw Industrial Editorial aesthetic utilizing Swiss 90-degree grid borders, deep matte black (#0A0A0A), monochrome 1px dividers, high-contrast Instrument Serif display typography, and JetBrains Mono micro-labels. This mirrors high-end physical SCADA monitoring terminals used in aerospace and cold chain infrastructure.

Q12: How is the platform deployed and hosted on Vercel?

The project is structured with a root vercel.json that directs Vercel's build pipeline: 'cd frontend && npm install && npm run build' with output directory 'frontend/dist'. We included an SPA rewrite rule (source: /(.*) -> /index.html) so client-side navigation never returns 404s. Additionally, we created api/index.py so FastAPI endpoints can execute as serverless Python functions on Vercel.

Q13: How does your system handle actuator automation (Compressors, Fans, Scrubbers)?

Each storage bin schema tracks physical actuator states: compressor_active (boolean), evaporator_fan_active (boolean), and ethylene_scrubber_active (boolean). When sensor telemetry detects ethylene exceeding the safety ceiling (e.g. >0.15 ppm), the system auto-activates the catalytic scrubber relay (120 m3/h bed) and logs a priority warning in the alarm journal. If cooling compressor fails, emergency backup alerts are triggered.

Q14: How can physical IoT hardware (ESP32 / Arduino / Raspberry Pi) be connected to your system?

Physical IoT sensor nodes installed inside chambers periodically assemble JSON telemetry packets containing bin_id, temperature, humidity, ethylene_ppm, and co2_ppm. They transmit these packets via HTTP POST or MQTT to our ingestion endpoint: POST /api/telemetry. The endpoint validates the payload with Pydantic, updates the digital twin state, appends to the historical time-series buffer, and triggers threshold checks.

Q15: What are the future research directions and enhancements for AgroTwin-Share?

1. Multi-gas integration (O2 and CO2 respiration quotient modeling for Controlled Atmosphere rooms).
2. Machine Learning residual calibration (using LSTM or XGBoost to correct biophysical Arrhenius predictions against empirical microbial plate counts).
3. Dynamic pricing integration: connecting produce SLI directly to wholesale commodity exchange APIs to automatically fire-sale produce with diminishing shelf life.

7. Quick-Reference Viva Cheatsheet

Memorize these core scientific equations, parameters, and architecture facts for instantaneous recall during viva questioning:

Concept / Parameter	Formula / Technical Value	Key Viva Explanation Point
Respiration Q10 Factor	f_T = Q10 ^ ((T - T_opt)/10) Apples: 2.2, Bananas: 2.6	Respiration roughly doubles every 10 deg C rise, consuming stored sugars and halving shelf-life.
Vapor Pressure Deficit (VPD)	VPD = VP_sat(T) x (1 - RH/100) Tetens formula for VP_sat	True drying gradient of air. Optimal is 0.05-0.25 kPa. High VPD causes shriveling; RH > 96% causes mold.
Chilling Injury	T < T_chill (Bananas < 12 deg C, Avocados < 3 deg C)	Membrane lipid phase transition from liquid to gel, causing permanent tissue collapse and decay.
Ethylene Sensitivity	Climacteric: Apples, Bananas, Avocados, Tomatoes	Autocatalytic surge triggers pectinase and cellulase enzyme synthesis, breaking cell walls.
Shelf-Life Index (SLI %)	SLI = 100% (Harvest) to 0% (Rot) Floor: 10% (Unmarketable)	Integrated decay equation tracking quality degradation. Translates directly into days remaining.
Service Architecture	const USE_MOCK = true; FastAPI: http://localhost:8000	Decoupled adapter pattern. Offline client simulation for expo + full-stack Python server for production.
Color Palette (60-30-10)	60%: #0A0A0A, 30%: #EDEDED / #262626 10%: #00FF66, #FFB000, #FF3333	Raw industrial terminal aesthetic with zero consumer rounded badges. Maximum clarity and contrast.

8. Final Words for the Team: How to Score Maximum Marks

1. **Speak with confidence about the biology:** Examiners respect computer science students who deeply understand the agricultural domain (Arrhenius kinetics, Tetens formula, climacteric curves).
2. **Emphasize economic value:** Always point to the *$193,000 biomass valuation* and show how 4 degrees of temperature deviation causes an $18,000 loss—this transforms an engineering project into a commercial business solution.
3. **Show off the UI live:** Don't just click through static screens; use the **[+C2H4 SPIKE]** and **[+DOOR LEAK]** buttons to prove the digital twin recalculates shelf-life in real time.
4. **Highlight Vercel and GitHub:** Emphasize that the codebase is completely open-source, version-controlled on GitHub (aaronaji007/AgroTwin), and deployed live on Vercel.